PENNSTATE

October 12, 2025

From: Siena Romano, Graduate Student Consultant

To: Eva Green, Researcher

Analysis of Covariance and Relationships Found Within Mutation Rates and Between Other Genomic Features

EXECUTIVE SUMMARY

In response to the research questions concerning the covariance and overall associations within four different human genome mutation rates and between these mutation rates and other genomic features, Penn State MAS graduate student consultants applied and generated the following tests and conclusions:

Principal Component Analysis (PCA) summarized which mutation rates significantly co-vary with each other. The results include a graph that depicts small insertion, small deletion, and nucleotide substitution rates most co-vary with each other and the microsatellite repeat number alteration rate is independent from the others.

Multivariate Analysis of Variance (MANOVA) compared mutation rates across the six states of genetic divergence and concluded that the average mutation rate depends on the genetic state. Additional testing using individual ANOVAs for each mutation rate confirmed that each of the four mutation rates' average rate differs in at least one of the six genetic states. Further, MANOVA contrasts comparing mutation rates of state 1 to state 6 confirmed that the mutation rates found in these specific states are significantly different on average.

Canonical Correlation Analysis (CCA) determined that small insertion, small deletion, and nucleotide substitution rates correlate and capture the most variation with the nCGm and LINE genomic features. Individual regressions with each mutation rate as the response also confirmed nCGm and LINE are significant predictors for small insertion, small deletion, and nucleotide substitution rates.

The key findings were the strong connection between the small insertion, small deletion, and nucleotide substitution rates themselves and with the nCGm and LINE features. Additionally, it is important to note the independence of the microsatellite repeat number alteration rate and that, on average, mutation rates from all genetic states (including state 1 and state 6) differ significantly.

1.0 - PROJECT DESCRIPTION

Researchers at Penn State's Huck Institute for Life Sciences are conducting an experimental study that collected 2,735 chromosome samples across six states of genetic variation. The researchers sampled each of the 23 chromosome pairs found in human DNA (including only the X chromosome pair for participants that are biologically female) and divided participants' human genomes into 1-Mb windows for a more manageable analysis. From the 23 chromosome sets, researchers extracted various genomic features and calculated mutation rates of small insertion, small deletion, nucleotide substitutions, and microsatellite repeat number alterations.

With the assistance of graduate student consultants, the researchers are in the analysis stage of the experimental study and have an objective to obtain graphical and statistical analyses on the co-variation between the four mutation rates, the difference in mutation rates found across six states of genetic divergence, and the relationship between the mutation rates and potential explanatory features. The conclusions found provide the scientific community with insights into the association of altered genome sequences across different states of genetic diversity and the relationships between those mutations and other genomic elements.

1.1 - RESEARCH QUESTIONS

Question 1: Do the four mutation rates co-vary with each other across the human genome? Can it be illustrated in a two-dimensional plot?

Question 2: Do our four mutation rates differ among the six states of genetic divergence identified in the human genome? Do the average mutation rates in state 1 significantly differ from those in state 6?

Question 3: Can we link mutation rates and their co-variation to the genomic features?

1.2 - VARIABLES

Please refer to Appendix Table 1 for the variables included in the humanAR_1mbData CSV and their descriptions, data types, and numeric ranges. As a note, mutation rates were standardized prior to the analyses so that mutation rates could be compared on the same scale.

2.0 - EXPLORATORY DATA ANALYSIS (EDA)

Numerical Summaries

Table 2 (Appendix) shows that as the data was standardized, the mean values are all near zero, with standard deviations close to one. The min and max range between roughly –4 to 6, suggesting there could possibly be some outliers, but nothing too extreme for the size of this dataset.

Table 3 (Appendix) shows that GC, NLp, SINE, and nCGm are approximately symmetric, with mean and median near each other. Several features display right-skewness. The CpG variable has a large difference between mean and median, with values of 9.1 and 5 respectively. The max of 175 indicates a long tail. SINE also shows some right skewness with a mean greater than the median.

Graphical Summaries

Figure 1 (Appendix) displays histograms of the standardized mutation rates. All mutation rates appear roughly normally distributed.

Figure 2 (Appendix) displays histograms of the genomic features. GC, LINE, nCGm appear roughly normally distributed. SINE appears to be right skewed, and CpG appears to have a long-tailed distribution.

Figure 3 (Appendix) displays standardized mutation rate distributions across the six states of genetic divergence using box and whisker plots to show a comparison of the middle 50% of rates across each state. Outliers were also noted but were not removed as the outliers did not impact the ANOVA results.

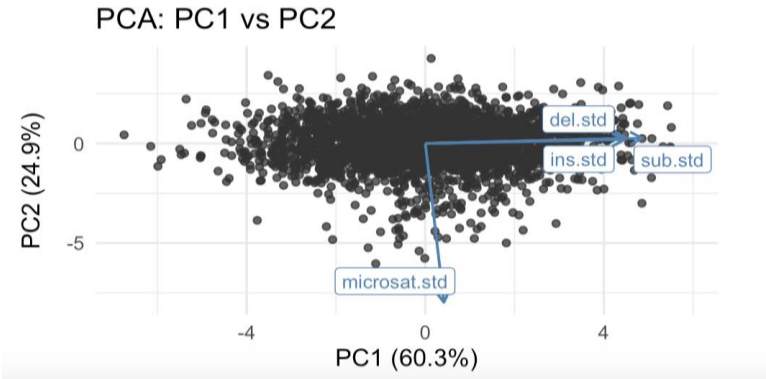
3.0 - STATISTICAL ANALYSIS

3.1 *- Do the four mutation rates co-vary with each other across the human genome? Can it be illustrated in a two-dimensional plot?*

PCA was performed to both analyze and depict how the four mutation rates vary with each other. PCA is a dimension reduction technique that reduces our data to a smaller number of components to summarize and visualize the relationship between mutation rates.

The scree plot (see Appendix Figure 4) shows the first two components contain over 85% of the variation in the mutation rates, with the first component alone containing 50% of it. Appendix Table 4 shows that the first principal component is made up of ins.std, del.std, and sub.std and the second is made up of microsat.std.

Based on the principal components and biplot included below, the mutation rates ins.std, del.std, and sub.std positively co-vary with each other.



Specifically, as PC1 increases, so do these three mutation rates. The second component is made up almost entirely of microsat.std, with a loading of nearly -1, showing it to be independent and does not co-vary with the other mutation rates. The correlations and scatterplots (see Appendix Figure 5) agree with the PCA, showing both the strong connection between ins.std, del.std, and sub.std, and the independence of microsat.std.

3.2 - *Do the four mutation rates differ among the six states of genetic divergence identified in the human genome? Do the average mutation rates in state 1 significantly differ from those in state* 6?

A one-way MANOVA was performed to determine if the four mutation rates differ in the six states of genomic divergence. MANOVA confirms if mutation rates depend on the genomic state by comparing the mean mutation rate vectors from each state. To conclude that there is a difference, at least one genomic state's mean mutation rate vector should be significantly different. Refer to Appendix Hypotheses & Assumptions for the MANOVA hypotheses.

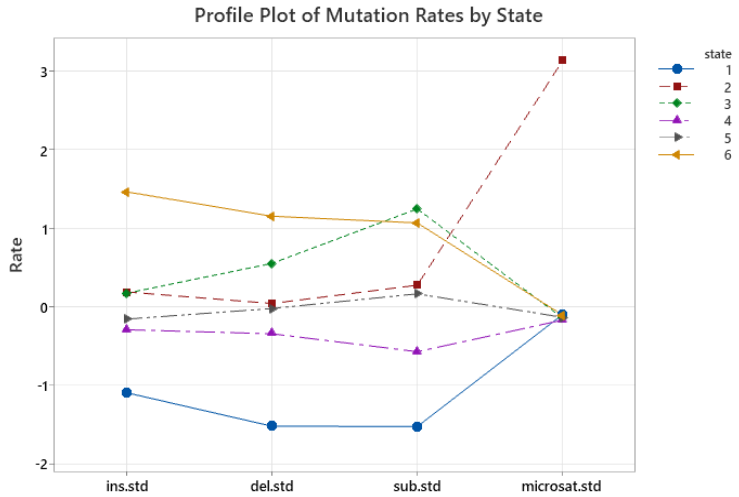
Before performing the analysis, all four MANOVA assumptions passed testing. Valid assumptions are critical to the integrity of the MANOVA test as assumptions that are violated can lead to misleading or inaccurate model outcomes. See Appendix Hypotheses & Assumptions for details on each assumption.

After confirming the assumptions were satisfied, the MANOVA test was performed, and the results are below.

MANOVA Test					
Statistic	Value	F Value	Num DF	Den DF	PR > F
Wilks' Lambda	0.37633076	154.92	20	9042.1	<0.0001

With a low p-value (the probability of obtaining these results if the null hypothesis were true), it can be concluded that the mean mutation rates depend on the genomic divergent state. Specifically, mean mutation rates differ for at least one pair of genomic states.

For further testing over the mean differences found in mutation rates across the six states of genetic divergence, refer to Appendix Table 5 for individual ANOVA test results and see the profile plot below for a visual representation.



Based on the ANOVA results tables, each mutation's test yielded a low p-value. Therefore, it can be concluded that each mean mutation rate is significantly different in at least one of the six states.

Last, MANOVA contrasts were used to determine whether the average mutation rate in state 1 and 6 differ. With a low p-value, it can be concluded that the states' mean mutation rates are significantly different.

MANOVA for States 1 vs 6					
Statistic	Value	F Value	Num DF	Den DF	PR > F
Wilks' Lambda	0.73065837	251.22	4	2726	<0.0001

3.3 - Can we link mutation rates and their co-variation to the genomic features?

CCA was used to examine the relationship between mutation rates and their co-variation with genomic features. CCA aims to find the strongest multivariate relationship between two sets of variables by creating pairs of linear combinations that are maximally correlated with each other. In this case, pairs of linear combinations from the mutation rates and genomic features were identified that maximize their correlations to determine which variables in each set are most strongly related to the others. First, graduate student consultants tested the statistical significance of the canonical correlation for each set of pairs. See Hypotheses & Assumptions for the CCA hypotheses and see below for the chart of the canonical correlations and their p-values.

Canonical	Correlation	Wilks' P-Value
CV1	0.2119	0.0000
CV2	0.0578	0.2969
CV3	0.0436	0.4101
CV4	0.0334	0.3842

The first pair of canonical correlations is significant, with a correlation of 0.2119, and a p-value of effectively 0. The next pair is not significant with a p-value of 0.2969. Considering the significance of the first pair, the loadings for each of the canonical variates were examined (see Appendix Tables 6 and 7). The loadings in the first mutation canonical variate show ins.std, del.std, and sub.std strongly contributing, while LINE and nCGm strongly contribute to the first genomic features canonical variate. Therefore, it can be concluded that ins.std, del.std, and sub.std are all associated with the nCGm and LINE genomic features.

Following CCA, graduate students conducted individual regression analyses on each mutation rate to see if any of the genomic features are significant predictors. The tables for the individual regressions of each mutation rate (see Appendix Table 8) show results consistent with the CCA output. Microsat.std does not have any significant predictors, whereas ins.std, del.std, and sub.std have LINE and nCGm as significant predictors. The residual plots and Q-Q plots (Appendix Figure 6) for ins.std, del.std, and sub.std show well-behaved residuals, supporting the regression models.

4.0 - RECOMMENDATIONS

Question 1: Based on the biplot of PCA loadings and co-variances found in Section 3.1, small insertion, small deletion, and nucleotide substitution rates positively co-vary and the microsatellite repeat number alterations rate is independent of the other three rates.

Question 2: Results from MANOVA and individual ANOVAs found in Section 3.2 confirm that the four mutation rates depend on the state of divergence and each mutation rate differs significantly in at least one of the six genetic states on average. Additionally, the mean mutation rate found in state 1 is significantly different from that found in state 6.

Question 3: Referring to the canonical correlation outputs in Section 3.3, nCGm and LINE are significantly related to small insertion, small deletion, and nucleotide substitution rates. Specifically, the genome mutations are positively correlated to LINE at 0.665 and negatively correlated to nCGm at -0.799 and the genome features are strongly and positively correlated to small insertion, small deletion, and nucleotide substitution rates at 0.946, 0.877, and 0.818. Additional regression testing confirmed that nCGm and LINE are significant predictors of small insertion, small deletion, and nucleotide substitution rates and the microsatellite repeat number alterations rate does not have a significant relationship with any of the genomic features.

5.0 - RESOURCES

Please refer to Appendix Table 9 for academic papers and related R or SAS tutorials for more information on the testing methods used in this report and their interpretations.

6.0 - CONSIDERATIONS

Graduate student consultants performing the analyses had little background on the experimental design used to collect the human genome data. Further details on the experimental study could provide evidence that sample data was collected randomly and independently.

Additionally, graduate student consultants were not advised on why a limited number of genome features were included in the data set. It is possible that there are other genome features that are significantly related to the four mutation rates, and readers should not conclude that the features listed are the only significant predictors.

Last, mutation rate outliers were noted after standardization. These outliers were not removed before performing the analyses as the data set was large and the outliers did not change the testing results of any of the research questions. Researchers should consider exploring the outliers identified in the graduate student consultants' figures.

APPENDIX

TABLES & FIGURES

Table 1: Data Set's Variables and their Descriptions, Data Types, and Range

Variable	Description	Type	Range/Notes
Chr	Chromosome number of 1Mb window	Integer	
Start	Window start position (bp)	Integer	
End	Window end position (bp)	Integer	
ins.std	Std. insertion rate	Continuous; response	Standardized; -4 - 4
del.std	Std. deletion rate	Continuous; response	Standardized; -4 - 4
sub.std	Std. substitution rate	Continuous; response	Standardized; -4 - 4
microsat.std	Std. microsatellite alterations	Continuous; response	Standardized; -4 - 6
GC	GC content	Continuous; explanatory	Proportion; 27 - 59
CpG	CpG island count	Integer; explanatory	0 - 175
nCGm	non-CpG methyl-cytosine count	Integer; explanatory	132 - 10863
LINE	LINE element count	Integer; explanatory	221 - 812
SINE	SINE element count	Integer; explanatory	128 - 2030
NLp	Nuclear lamina regions	Integer; explanatory	9 - 1062
State	Genetic divergence state	Categorical; explanatory	1, 2, …6

Table 2: Summary Statistics of Standardized Mutation Rates

	Variable	Mean	Median	SD	Min	Max	Missing
ins.std	ins.std	-0.0042519	-0.0669127	1.178071	-4.076456	4.182735	0
del.std	del.std	-0.0199801	0.0028275	1.145401	-4.217459	3.907991	0
sub.std	sub.std	0.0099429	0.0328628	1.251695	-4.312136	4.187625	0
microsat.std	microsat.std	-0.0159831	-0.0791329	1.193353	-4.272477	5.961870	0

Table 3: Summary Statistics of Genomic Features

	Variable	Mean	Median	SD	Min	Max	Missing
GC	GC	40.684854	40.0076	4.342045	27.7245	58.8723	0
CpG	CpG	9.146618	5.0000	13.501120	0.0000	175.0000	0
nCGm	nCGm	5449.617550	5482.0000	1410.411525	132.0000	10863.0000	0
LINE	LINE	490.685923	487.0000	73.234427	221.0000	812.0000	0
SINE	SINE	623.266910	544.0000	315.115482	128.0000	2030.0000	0
NLp	NLp	515.530530	543.0000	262.219994	9.0000	1062.0000	0

Table 4: PCA Loadings for Mutation Rates (PC1 & PC2)

Variable	PC1 Loading	PC2 Loading
ins.std	0.556	0.024
del.std	0.564	0.033
sub.std	0.607	0.033
microsat.std	0.052	-0.999

Table 5: Individual ANOVAs for Mutation Rates

Dependent Variable: insStd

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1078.792499	215.758500	216.82	<.0001
Error	2729	2715.590048	0.995086		
Corrected Total	2734	3794.382547			

Dependent Variable: delStd

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1027.079415	205.415883	219.00	<.0001
Error	2729	2559.775941	0.937990		
Corrected Total	2734	3586.855356			

Dependent Variable: subStd

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1489.417145	297.883429	290.95	<.0001
Error	2729	2794.052637	1.023838		
Corrected Total	2734	4283.469782			

Dependent Variable: microsatStd					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1085.375754	217.075151	210.96	<.0001
Error	2729	2808.089278	1.028981		
Corrected Total	2734	3893.465032			

Table 6: Loadings of Mutation Rates with Canonical Variates

Mutation	U1	U2	U3	U4
ins.std	0.9461731	0.2933431	0.1367459	0.0026077
del.std	0.8771509	-0.3923745	0.1073043	-0.2552143
sub.std	0.8177016	-0.3697848	-0.1589872	0.4115172
microsat.std	0.1286335	0.1663830	-0.9104989	-0.3560363

Table 7: Loadings of Genomic Features with Canonical Variates

Genomic	V1	V2	V3	V4	V5	V6
GC	-0.3966014	0.5779116	0.1056483	0.3880609	0.1461482	0.5706255
CpG	-0.3246485	0.1830319	0.5886000	0.1086565	0.3589670	0.6115466
nCGm	-0.7987977	-0.2940728	-0.3741971	0.1531132	-0.1970609	0.2704501
LINE	0.6649031	-0.0987950	-0.3602179	-0.0657794	-0.3088061	0.5645337
SINE	-0.4471601	0.3745721	0.2552429	0.2476193	-0.5145048	0.5182317
NLp	0.1137271	0.1835859	-0.3730708	-0.8465520	0.1736047	-0.2595991

Table 8: Regression Coefficients by Mutation Rate

Regression Coefficients for sub.std

term	estimate	std.error	statistic	p.value
GC	0.000	0.010	-0.021	9.83e-01
CpG	-0.002	0.003	-0.807	4.20e-01
nCGm	0.000	0.000	-4.401	1.12e-05
LINE	0.002	0.000	5.031	5.00e-07
SINE	0.000	0.000	-1.971	4.88e-02
NLp	0.000	0.000	-2.820	4.84e-03

Regression Coefficients for del.std

term	estimate	std.error	statistic	p.value
GC	-0.008	0.009	-0.900	3.68e-01
CpG	0.000	0.003	0.001	9.99e-01
nCGm	0.000	0.000	-4.987	7.00e-07
LINE	0.002	0.000	5.277	1.00e-07
SINE	0.000	0.000	-1.474	1.41e-01
NLp	0.000	0.000	-2.062	3.93e-02

Regression Coefficients for microsat.std

term	estimate	std.error	statistic	p.value
GC	0.011	0.010	1.147	0.251
CpG	-0.003	0.003	-1.187	0.235
nCGm	0.000	0.000	-0.042	0.966
LINE	0.000	0.000	1.369	0.171
SINE	0.000	0.000	-0.729	0.466
NLp	0.000	0.000	0.845	0.398

Regression Coefficients for ins.std

term	estimate	std.error	statistic	p.value
GC	0.005	0.010	0.558	5.77e-01
CpG	-0.003	0.003	-1.138	2.55e-01
nCGm	0.000	0.000	-6.727	0.00e+00
LINE	0.002	0.000	5.111	3.00e-07
SINE	0.000	0.000	-1.178	2.39e-01
NLp	0.000	0.000	-1.826	6.79e-02

Table 9: Related Academic Papers and R or SAS Tutorials

Principal Component Analysis	• *Principal component analysis: a review and recent developments* by Ian T Jolliffe and Jorge Cadima • "Principal components" from UCLA's Statistical Consulting Group
Multivariate Analysis of Variance	• *Reporting and Interpreting Multivariate Analysis of Variance (MANOVA): Adopting the Best Practices in Educational Research* by Simon Ntumi • "One-way MANOVA | SAS Data Analysis Examples" from UCLA's Statistical Consulting Group

Canonical Correlation Analysis	• *A Demonstration of Canonical Correlation Analysis with Orthogonal Rotation to Facilitate Interpretation* by Patrick V. Dattalo • "Canonical Correlation Analysis | R Data Analysis Examples" from UCLA's Statistical Consulting Group

Figure 1: Histograms for Mutation Rates

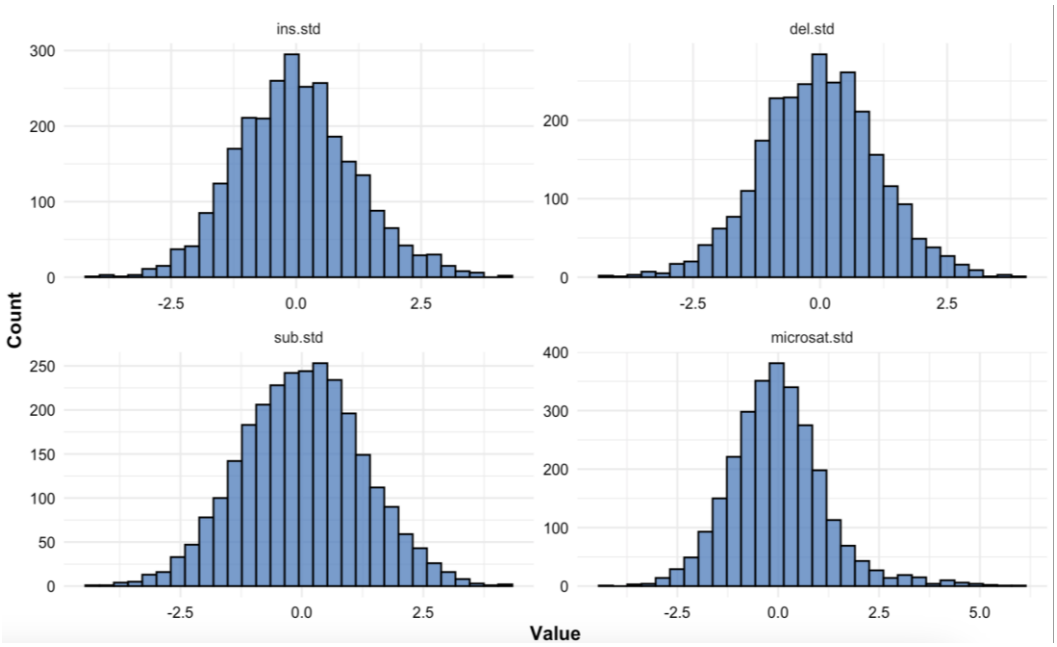


Figure 2: Histograms for Genomic Features

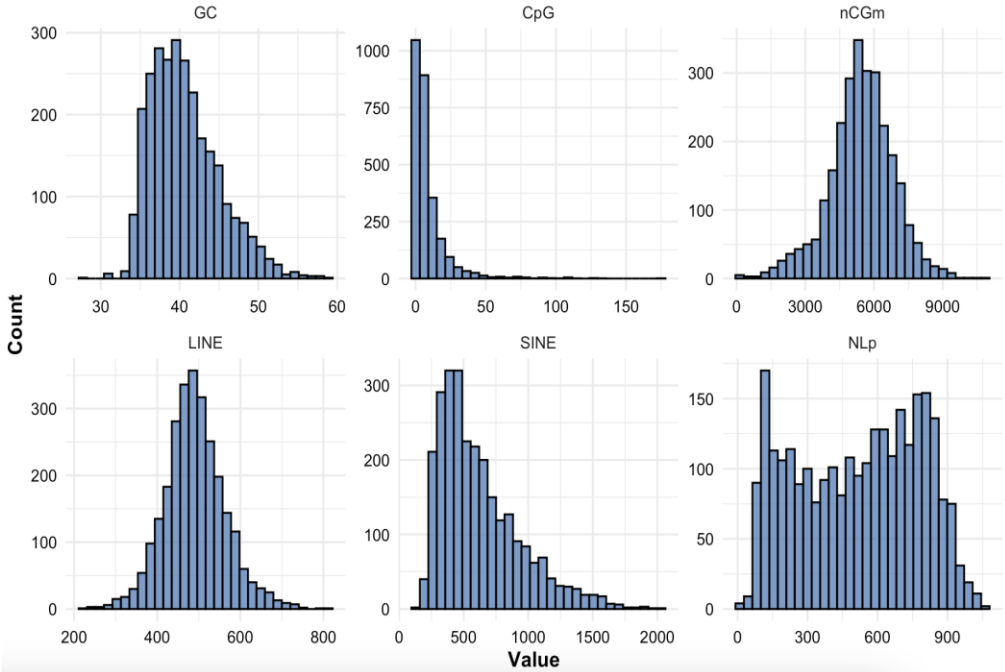
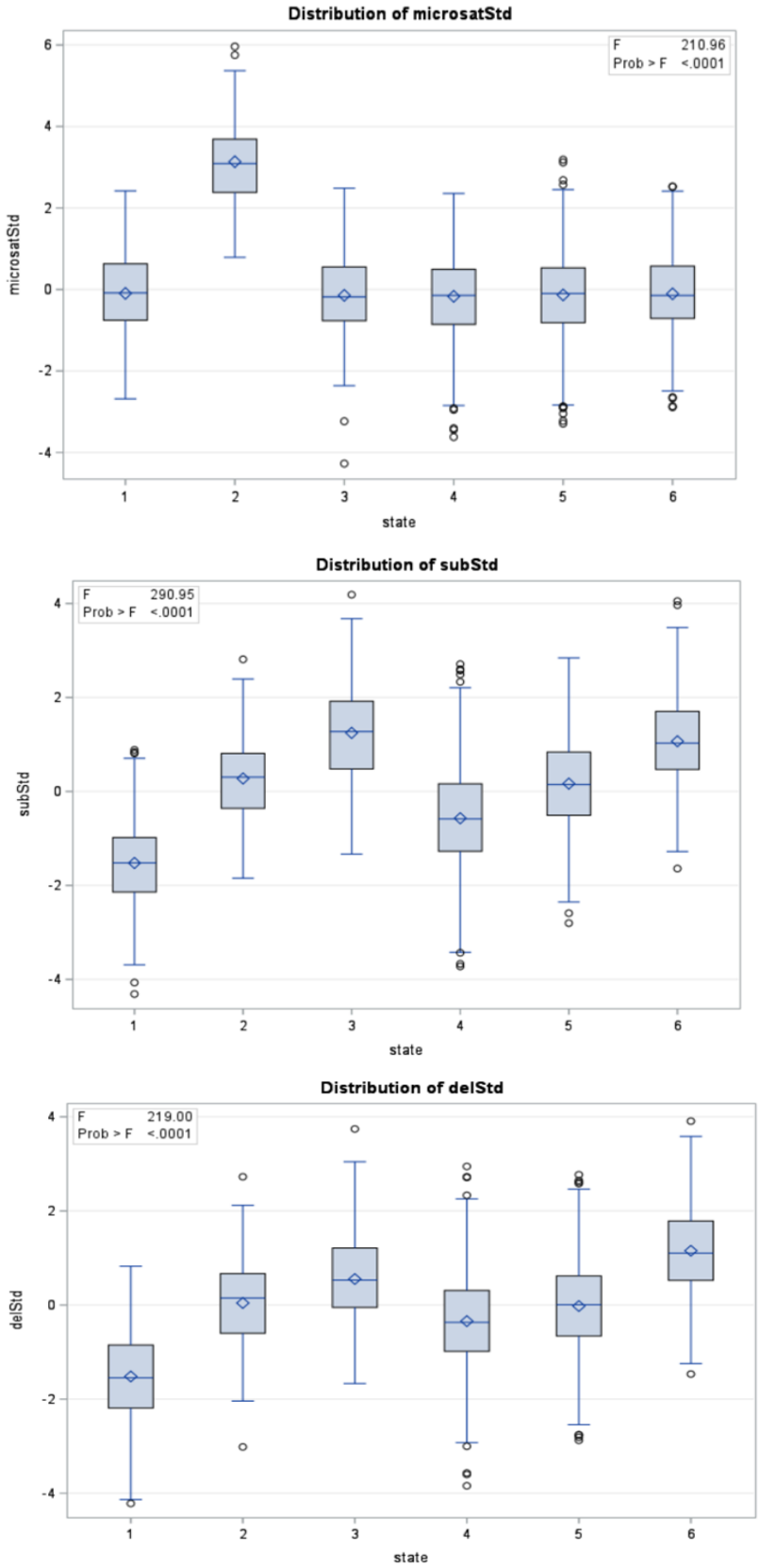


Figure 3: Box and Whisker Plots for Mutation Rates Across all Six States of Divergence



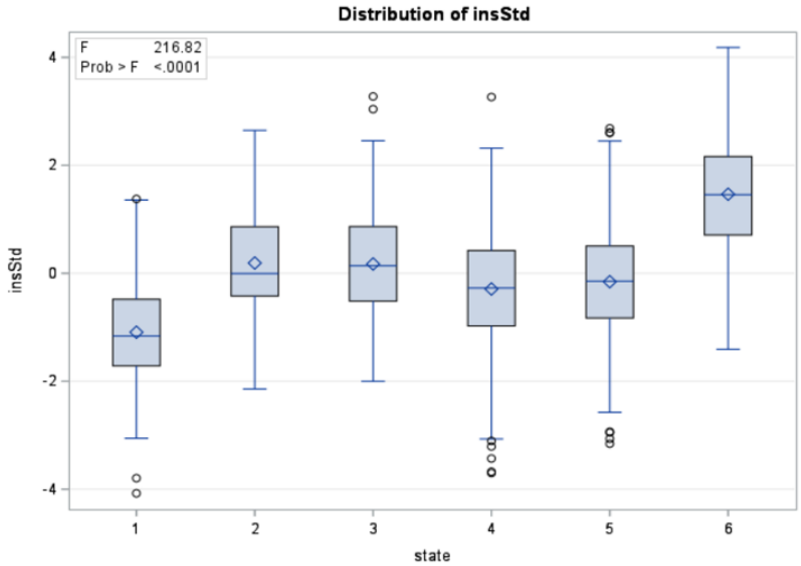


Figure 4: Scree Plot of PCA Loadings

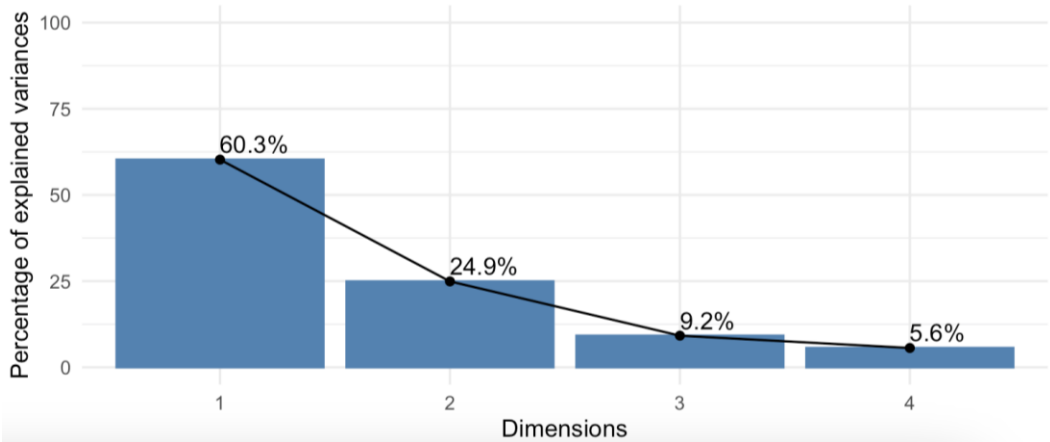


Figure 5: Correlation Table for Mutation Rates with Histogram and Scatter Plots

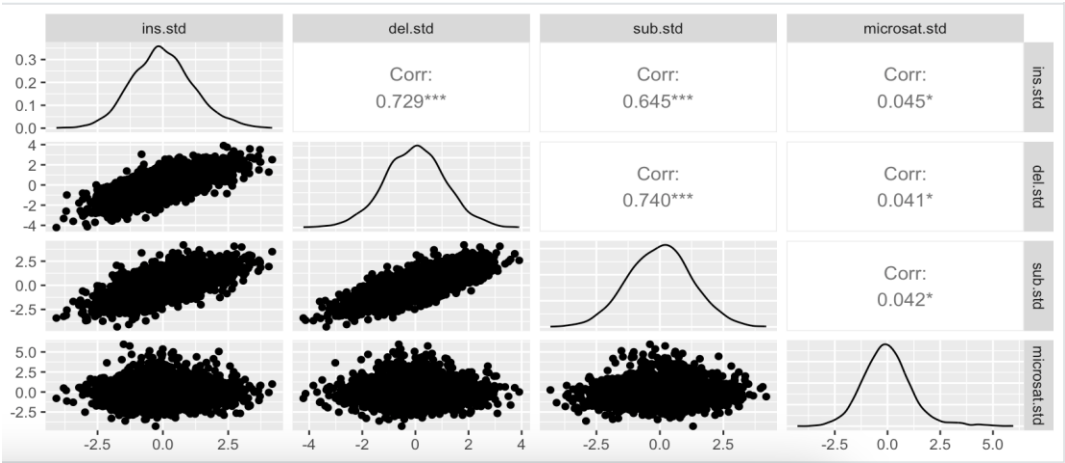
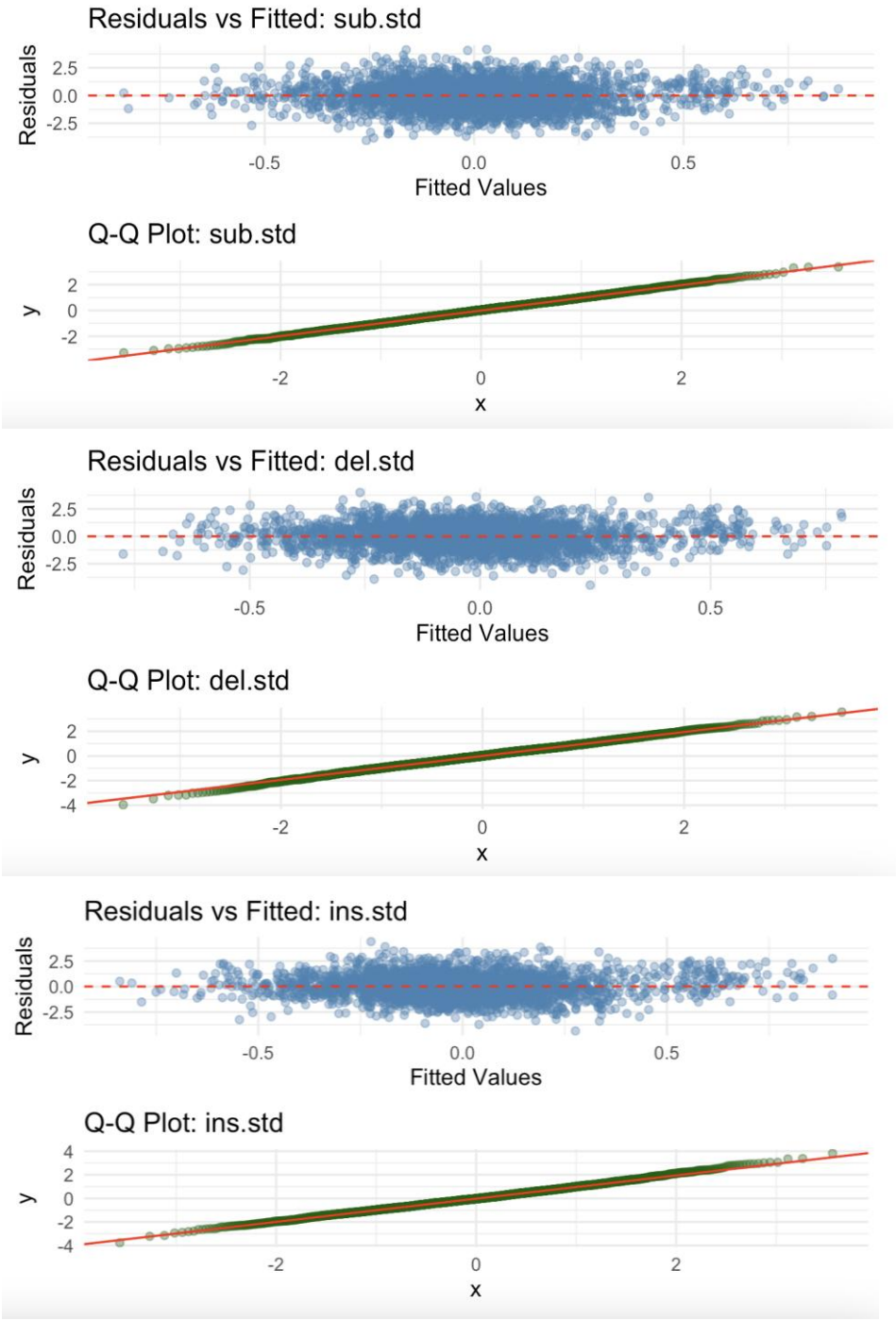


Figure 6: Residual plots for Individual Mutation Rates Regressions



Hypotheses & Assumptions

MANOVA Hypotheses

H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu_6$

H_a: $\mu_{ik} \neq \mu_{jk}$ *for at least one* $i \neq j$ *and at least one variable* k

where μ_{ik} *is the genomic state mean vector for variable* k

CCA Hypotheses

H_0: $\beta_{ij} = 0$ for i = 1, 2, 3, 4, 5, 6, 7 and j = 1, 2, 3, 4

H_a: $\beta_{ij} \neq 0$ for i = 1, 2, 3, 4, 5, 6, 7 and j = 1, 2, 3, 4

Where β_{ij} is the regression coefficients for genome features (i) and genome mutation rates (j)

MANOVA Assumptions

1. The assumption that the mutation rates from the six genomic states have common mean vectors is satisfied as there is no evidence to suggest that this isn't a randomized experiment and that there would be different mean mutation rate results from another population sample. Passing the assumption that there is equal mean variances within each mutation rate group ensures that if there are significant MANOVA results, it is due to mean differences across groups and not within.
2. The assumption that the mutation rates in all states have a common variance-covariance matrix is satisfied according to Box's test. Specifically, the high p-value of 0.229 indicates that the chi-square value is not significant. Therefore, there is no statistically significant evidence that the variance-covariance matrices are not homogenous.

Box's Test Hypotheses:

H_0: $\Sigma_1 = \Sigma_2 = \Sigma_3 = \Sigma_4 = \Sigma_5 = \Sigma_6$

H_a: $\Sigma_i \neq \Sigma_j$ *for at least one* $i \neq j$

where Σ_i *is the covariance matrix of the genomic state*

Box's Test Results:

Chi-Square	DF	P-Value
57.102653	50	0.2281

The assumption that there are common variance-covariance matrices between mutation rates across the six states confirms that the groups' data points are similar enough in spread to perform MANOVA.

3. The Independence assumption is satisfied as there is no evidence to suggest that chromosome data was not independently sampled in the study. Passing this assumption is important to MANOVA as dependent samples can lead to inaccurate and biased results when testing if there are mean differences across groups.

4. The Normality assumption is satisfied by the Central Limit Theorem because of the large number of samples. Further, the mutations' histograms fall into an approximately symmetric bell shape. See Appendix Figure 1 for the mutation rate histograms. This assumption is important to performing MANOVA as the MANOVA model requires data to follow a central tendency and now be skewed in one direction.

CODE

R CODE FOR SUMMARY:

```
df <- read.csv("humanAR_1mbData.csv", header = TRUE)

library(knitr)
library(kableExtra)

num_summary <- function(df, vars){
  data.frame(
    Variable = vars,
    Mean = sapply(df[, vars], mean, na.rm = TRUE),
    Median = sapply(df[, vars], median, na.rm = TRUE),
    SD = sapply(df[, vars], sd, na.rm = TRUE),
    Min = sapply(df[, vars], min, na.rm = TRUE),
    Max = sapply(df[, vars], max, na.rm = TRUE),
    Missing = sapply(df[, vars], function(x) sum(is.na(x)))
  )
}

mutation_vars <- c("ins.std", "del.std", "sub.std", "microsat.std")
genomic_vars  <- c("GC", "CpG", "nCGm", "LINE", "SINE", "NLp")

summary_mutation <- num_summary(df, mutation_vars)
summary_genomic  <- num_summary(df, genomic_vars)

kable(summary_mutation, caption = "Table 1: Summary of standardized mutation rates", booktabs = TRUE) %>%
  kable_styling(full_width = FALSE, position = "center", stripe_color = "gray!15") %>%
  row_spec(0, bold = TRUE, color = "white", background = "#4F81BD")

kable(summary_genomic, caption = "Table 2: Summary of genomic features", booktabs = TRUE) %>%
  kable_styling(full_width = FALSE, position = "center", stripe_color = "gray!10") %>%
```

```
  row_spec(0, bold = TRUE, color = "white", background = "#4F81BD")

library(ggplot2)
library(reshape2)

df_mut_melt <- melt(df[, mutation_vars])

p_mut <- ggplot(df_mut_melt, aes(x = value)) +
  geom_histogram(bins = 30, fill = "#4F81BD", color = "black", alpha = 0.8) +
  facet_wrap(~ variable, scales = "free", ncol = 2) +
  labs(
    title = "Figure 1: Distribution of standardized mutation rates",
    x = "Value",
    y = "Count"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "black")
  )

print(p_mut)

df_gen_melt <- melt(df[, genomic_vars])

p_gen <- ggplot(df_gen_melt, aes(x = value)) +
  geom_histogram(bins = 30, fill = "#9BBB59", color = "black", alpha = 0.8) +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  labs(
    title = "Figure 2: Distribution of genomic features",
    x = "Value",
    y = "Count"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 16),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "black")
  )

print(p_gen)
print(p_mut)
```

R **CODE FOR ANALYSIS QUESTION 1***:*

```
mut_mat <- df %>%
  select(ins.std, del.std, sub.std, microsat.std) %>%
  drop_na() %>%
  as.matrix()
complete_idx <- complete.cases(mut_mat)
mut_mat <- mut_mat[complete_idx, , drop = FALSE]
df_sub <- df %>% slice(which(complete_idx))
pca_res <- prcomp(mut_mat, center = TRUE, scale. = FALSE)
eig_vals <- (pca_res$sdev)^2
var_explained <- eig_vals / sum(eig_vals)
p_scree <- fviz_eig(pca_res, addlabels = TRUE, ylim = c(0, 100)) +
  ggtitle("Scree plot — Variance explained") +
  theme_minimal(base_size = 13)
scores <- as.data.frame(pca_res$x) %>%
  rownames_to_column(var = "window_id")
p_scatter <- ggplot(scores, aes(x = PC1, y = PC2)) +
  geom_point(size = 1.5, alpha = 0.8, color = "grey20") +
  labs(
    x = paste0("PC1 (", round(100 * var_explained[1], 1), "%)"),
    y = paste0("PC2 (", round(100 * var_explained[2], 1), "%)"),
    title = "PCA: PC1 vs PC2"
  ) +
  theme_minimal(base_size = 13)
loadings <- as.data.frame(pca_res$rotation[, 1:2]) %>%
  rownames_to_column(var = "variable")
multiplier <- min(
  (max(scores$PC1) - min(scores$PC1)) / (max(loadings$PC1) - min(loadings$PC1)),
  (max(scores$PC2) - min(scores$PC2)) / (max(loadings$PC2) - min(loadings$PC2))
) * 0.8
loadings_scaled <- loadings %>%
  mutate(PC1s = PC1 * multiplier,
         PC2s = PC2 * multiplier)
p_biplot <- p_scatter +
  geom_segment(data = loadings_scaled,
               aes(x = 0, y = 0, xend = PC1s, yend = PC2s),
               arrow = arrow(length = unit(0.25, "cm")), linewidth = 0.7,
               color = "steelblue") +
  geom_label_repel(data = loadings_scaled,
                   aes(x = PC1s, y = PC2s, label = variable),
                   size = 3.6, color = "steelblue", fill = "white") +
  theme(legend.position = "none")
p_contrib <- fviz_contrib(pca_res, choice = "var", axes = 1:2, top = 10) +
  ggtitle("Variable contributions to PCs") +
  theme_minimal(base_size = 13)
```

```
pca_res <- prcomp(mut_mat, center = TRUE, scale. = FALSE)
loadings_tbl <- as.data.frame(pca_res$rotation[, 1:2]) %>%
  rownames_to_column("Variable") %>%
  rename(
    "PC1 Loading" = PC1,
    "PC2 Loading" = PC2
  ) %>%
  mutate(across(-Variable, ~ round(.x, 3)))
loadings_tbl %>%
  kable("html", caption = "PCA Loadings for Mutation Rates (PC1 & PC2)") %>%
  kable_styling(full_width = FALSE, position = "center", bootstrap_options = c("striped", “hover"))
print(p_scree)
print(p_biplot)
```

SAS Code for Analysis Question 2:

Box’s Test:

```
options ls=78;
title "Box's Test - Genome Mutations by State";

data mutations;
infile "\Users\siena\OneDrive\Documents\STAT580\humanAR_1mbData.csv" firstobs=2 delimiter=",";
input chr $ start end state $ GC CpG nCGm LINE SINE NLp insStd delStd subStd microsatStd;
run;

proc discrim data=mutations pool=test SLPOOL=0.05;
  class state;
  var insStd delStd subStd microsatStd;
  run;
```

MANOVA, Individual ANOVAs, and Box and Whisker Plots:

```
proc glm data=mutations alpha=0.05;
  class state;
  model insStd delStd subStd microsatStd = state;
  contrast '1 vs 6' state  1 0 0 0 0 -1;
  estimate '1 vs 6' state  1 0 0 0 0 -1;
  lsmeans state;
  lsmeans state / stderr;
  manova h=state / printe printh;
  run;
```

R CODE FOR ANALYSIS QUESTION 3:

```
library(CCA)
library(CCP)
library(ggplot2)
```

```
library(ggpubr)
library(dplyr)
library(tibble)
library(knitr)
library(kableExtra)
library(scales)

mutation_vars <- c("ins.std", "del.std", "sub.std", "microsat.std")
genomic_vars  <- c("GC", "CpG", "nCGm", "LINE", "SINE", "NLp")

X <- as.matrix(na.omit(df[, mutation_vars]))
Y_all <- as.matrix(na.omit(df[, genomic_vars]))
complete_idx <- complete.cases(df[, c(mutation_vars, genomic_vars)])
X <- as.matrix(df[complete_idx, mutation_vars])
Y <- as.matrix(df[complete_idx, genomic_vars])
n <- nrow(X)
p <- ncol(X)
q <- ncol(Y)
cca_res <- cancor(X, Y)
can_corrs <- cca_res$cor
can_corrs_rounded <- round(can_corrs, 3)


sig_test <- p.asym(can_corrs, n, p, q, tstat = "Wilks")
sig_table <- data.frame(
  Canonical = paste0("CV", seq_along(can_corrs)),
  Correlation = round(can_corrs, 4),
  Wilks_p = signif(sig_test$p.value, 4)
)
kable(sig_table, caption = "Table: Canonical correlations and Wilks' Lambda p-values") %>%
  kable_styling(full_width = FALSE)
X_scores <- scale(X, center = TRUE, scale = FALSE) %*% cca_res$xcoef
Y_scores <- scale(Y, center = TRUE, scale = FALSE) %*% cca_res$ycoef

colnames(X_scores) <- paste0("U", seq_len(ncol(X_scores)))
colnames(Y_scores) <- paste0("V", seq_len(ncol(Y_scores)))

loadings_X <- cor(X, X_scores)
loadings_Y <- cor(Y, Y_scores)

loadings_X_df <- as.data.frame(loadings_X) %>% rownames_to_column("Mutation")
loadings_Y_df <- as.data.frame(loadings_Y) %>% rownames_to_column("Genomic")

kable(loadings_X_df %>% select(Mutation, starts_with("V") | starts_with("U")) %>% head(., nrow(loadings_X_df)),
    caption = "Table: Loadings (correlations) of mutation rates with canonical variates") %>%
  kable_styling(full_width = FALSE)
```

```
 kable(loadings_Y_df %>% select(Genomic, starts_with("V") | starts_with("U")) %>% head(., nrow(loadings_Y_df)),
      caption = "Table: Loadings (correlations) of genomic features with canonical variates") %>%

kable_styling(full_width = FALSE)

library(tidyverse)
library(patchwork)

mutations <- c("ins.std", "del.std", "sub.std", "microsat.std")
features <- c("GC", "CpG", "nCGm", "LINE", "SINE", "NLp")
models <- list()
resid_plots <- list()
for (mut in mutations) {

  formula <- as.formula(paste(mut, "~", paste(features, collapse = " +  ")))

  models[[mut]] <- lm(formula, data = df)

  p1 <- ggplot(models[[mut]], aes(.fitted, .resid))
    geom_point(alpha = 0.4, color = "steelblue")
    geom_hline(yintercept = 0, linetype = "dashed", color = "red")
    labs(title = paste("Residuals vs Fitted:", mut),
         x = "Fitted Values", y = "Residuals")
    theme_minimal(base_size = 14)

  p2 <- ggplot(models[[mut]], aes(sample = .stdresid))
    stat_qq(alpha = 0.4, color = "darkgreen")
    stat_qq_line(color = "red")
    labs(title = paste("Q-Q Plot:", mut))
    theme_minimal(base_size = 14)

  resid_plots[[mut]] <- p1 / p2
}
for (mut in mutations) {
  print(resid_plots[[mut]])
}
library(broom)

for (mut in names(models)) {

  coef_table <- tidy(models[[mut]]) %>%
    filter(term != "(Intercept)") %>%  # remove intercept
    mutate(
      estimate = round(estimate, 3),
      std.error = round(std.error, 3),
```

```
    statistic = round(statistic, 3),
    p.value = signif(p.value, 3)
  )

 cat("\n\n### Regression Table for", mut, "\n\n")
 print(
  coef_table %>%
    kable(format = "html", caption = paste("Regression Coefficients for", mut)) %>%
    kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "condensed"))
 )
}
```

CITATIONS

Dattalo, Patrick V. "A Demonstration of Canonical Correlation Analysis with Orthogonal Rotation to Facilitate Interpretation." *Social Work Publications*, Virginia Commonwealth University, 2014, scholarscompass.vcu.edu/cgi/viewcontent.cgi?article=1001&context=socialwork_pubs.

Jolliffe, Ian T, and Jorge Cadima. "Principal Component Analysis: A Review and Recent Developments." *Mathematical, Physical, and Engineering Sciences*, The Royal Society Publishing, 13 Apr. 2016, royalsocietypublishing.org/doi/10.1098/rsta.2015.0202.

Ntumi, Simon. "Reporting and interpreting multivariate analysis of variance (MANOVA): Adopting the best practices in educational research." *Journal of Research in Educational Sciences*, vol. 12, no. 14, 31 Dec. 2021, p. 48, https://doi.org/10.14505/jres.v12.14.05.

UCLA: Statistical Consulting Group. "CANONICAL CORRELATION ANALYSIS | R DATA ANALYSIS EXAMPLES." *OARC Stats*, UCLA: Advanced Research Computing, stats.oarc.ucla.edu/r/dae/canonical-correlation-analysis/. Accessed 4 Oct. 2025.

UCLA: Statistical Consulting Group. "ONE-WAY MANOVA | SAS DATA ANALYSIS EXAMPLES." *OARC Stats*, UCLA: Advanced Research Computing, stats.oarc.ucla.edu/sas/dae/one-way-manova/. Accessed 4 Oct. 2025.

UCLA: Statistical Consulting Group. "Principal Components." *OARC Stats*, UCLA: Advanced Research Computing, www.stat.ucla.edu/~rgould/252w02/principalcomp.pdf. Accessed 4 Oct. 2025.